M3 · Prompt–region alignment

Build image-specific prompts and ROI links

Image I

UPT detector

Object / relation phrases

CLIP text encoder

Text embeddings to, tr

UPT boxes / classes

YOLO boxes B

Class + IoU ≥ 0.5 matching

Prompt package
to, tr embeddings
ROI links + box geometry

Feeds both updates

Object prompt update

Vo (query + residual)

to + original Vo[i] + box geometry

Slot MLP

Slots Zo

K,V

Masked cross-attention

Projection

Vo′

Only slots linked to ROI i

Relation prompt update

Vr (query + residual)

tr + original Vo[s], Vo[o] + pair geometry

Slot MLP

Slots Zr

K,V

Masked cross-attention

Projection

Vr′

Only slots linked to s or o